

FSUJ6610

## Potassium hydroxide 1M alcoholic

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

产品说明: 氢氧化钾1M醇溶液 (5.6%氢氧化钾, 84.4%乙醇, 3.5%甲醇, 6.5%水)  
 Product Description: Potassium hydroxide 1M alcoholic

Cat No. : J/6610/15, J/6610/17, J/6610/24

Supplier **UK entity/business name**  
 Fisher Scientific UK  
 Bishop Meadow Road, Loughborough,  
 Leicestershire LE11 5RG, United Kingdom

**EU entity/business name**  
 Thermo Fisher Scientific  
 Janssen Pharmaceuticaaan 3a  
 2440 Geel, Belgium

Emergency Telephone Number Tel: 01509 231166  
 Chemtrec US: (800) 424-9300  
 Chemtrec EU: 001-703-527-3887

E-mail address begel.sdsdesk@thermofisher.com

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
 Liquid

**Appearance**  
 Colorless - Light yellow

**Odor**  
 No information available

#### Emergency Overview

Highly flammable liquid and vapor. Causes severe skin burns and eye damage. Harmful if swallowed. Harmful in contact with skin.  
 Harmful if inhaled. Causes damage to organs.

#### Classification of the substance or mixture

|                                                    |              |
|----------------------------------------------------|--------------|
| Flammable liquids.                                 | Category 2   |
| Acute Oral Toxicity                                | Category 4   |
| Acute Dermal Toxicity                              | Category 4   |
| Acute Inhalation Toxicity - Vapors                 | Category 4   |
| Skin Corrosion/Irritation                          | Category 1 A |
| Serious Eye Damage/Eye Irritation                  | Category 1   |
| Specific target organ toxicity - (single exposure) | Category 1   |

#### Label Elements



Signal Word

Danger

**Hazard Statements**

H225 - Highly flammable liquid and vapor  
H314 - Causes severe skin burns and eye damage  
H370 - Causes damage to organs  
H302 + H312 + H332 - Harmful if swallowed, in contact with skin or if inhaled

**Precautionary Statements****Prevention**

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P240 - Ground and bond container and receiving equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

Highly flammable liquid and vapor. Vapors may cause flash fire or explosion.

**Health Hazards**

Corrosive. Causes skin and eye burns. Harmful by inhalation, in contact with skin and if swallowed. Causes damage to organs.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. . Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component           | CAS No    | Weight % |
|---------------------|-----------|----------|
| Ethyl alcohol       | 64-17-5   | 84.4     |
| Potassium hydroxide | 1310-58-3 | 5.6      |
| Methyl alcohol      | 67-56-1   | 3.54     |
| Water               | 7732-18-5 | 6.46     |

**SECTION 4. FIRST AID MEASURES****Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required. Do not rub affected area.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

**Inhalation**

Remove to fresh air. If breathing is difficult, give oxygen. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

**Most important symptoms and effects**

Difficulty in breathing. Causes burns by all exposure routes. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically. Symptoms may be delayed.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water spray. Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

Do not use a solid water stream as it may scatter and spread fire.

**Specific Hazards Arising from the Chemical**

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Should not be released into the environment. Do not flush into surface water or sanitary sewer system. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

**Potassium hydroxide 1M alcoholic**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Use spark-proof tools and explosion-proof equipment. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance. Take precautionary measures against static discharges. Wash hands before breaks and immediately after handling the product. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Flammables area. Keep away from heat, sparks and flame.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters**

| Component           | China                                                           | Taiwan                                       | Thailand                     | Hong Kong                                                                                  |
|---------------------|-----------------------------------------------------------------|----------------------------------------------|------------------------------|--------------------------------------------------------------------------------------------|
| Ethyl alcohol       | -                                                               | TWA: 1000 ppm<br>TWA: 1880 mg/m <sup>3</sup> | TWA: 1000 ppm                | TWA: 1000 ppm<br>TWA: 1880 mg/m <sup>3</sup>                                               |
| Potassium hydroxide | Ceiling: 2 mg/m <sup>3</sup>                                    | -                                            | Ceiling: 2 mg/m <sup>3</sup> | Ceiling: 2 mg/m <sup>3</sup>                                                               |
| Methyl alcohol      | TWA: 25 mg/m <sup>3</sup><br>STEL: 50 mg/m <sup>3</sup><br>Skin | TWA: 200 ppm<br>TWA: 262 mg/m <sup>3</sup>   |                              | TWA: 200 ppm<br>TWA: 262 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 328 mg/m <sup>3</sup> |

| Component           | ACGIH TLV                             | OSHA PEL                                                                                                                                                                                 | NIOSH                                                                                                        | The United Kingdom                                                                                              | European Union                                               |
|---------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Ethyl alcohol       | STEL: 1000 ppm                        | (Vacated) TWA: 1000 ppm<br>(Vacated) TWA: 1900 mg/m <sup>3</sup><br>TWA: 1000 ppm<br>TWA: 1900 mg/m <sup>3</sup>                                                                         | IDLH: 3300 ppm<br>TWA: 1000 ppm<br>TWA: 1900 mg/m <sup>3</sup>                                               | TWA: 1000 ppm TWA;<br>1920 mg/m <sup>3</sup> TWA<br>WEL - STEL: 3000 ppm STEL; 5760 mg/m <sup>3</sup> STEL      |                                                              |
| Potassium hydroxide | Ceiling: 2 mg/m <sup>3</sup>          | (Vacated) Ceiling: 2 mg/m <sup>3</sup>                                                                                                                                                   | Ceiling: 2 mg/m <sup>3</sup>                                                                                 | WEL - 2 mg/m <sup>3</sup> STEL                                                                                  |                                                              |
| Methyl alcohol      | TWA: 200 ppm<br>STEL: 250 ppm<br>Skin | (Vacated) TWA: 200 ppm<br>(Vacated) TWA: 260 mg/m <sup>3</sup><br>(Vacated) STEL: 250 ppm<br>(Vacated) STEL: 325 mg/m <sup>3</sup><br>Skin<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup> | IDLH: 6000 ppm<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 325 mg/m <sup>3</sup> | WEL - TWA: 200 ppm<br>TWA: 266 mg/m <sup>3</sup> TWA<br>WEL - STEL: 250 ppm<br>STEL; 333 mg/m <sup>3</sup> STEL | TWA: 200 ppm 8 hr<br>TWA: 260 mg/m <sup>3</sup> 8 hr<br>Skin |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS70 General methods for sampling airborne gases and vapours MDHS 88

Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

## Exposure Controls

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

### Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Viton (R)      | > 480 minutes     | 0.7mm           | EN 374      | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure; Impervious clothing

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** low boiling organic solvent Type AX Brown conforming to EN371 or Organic gases and vapours filter Type A Brown conforming to EN14387

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141; Type AX  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

**Appearance** Colorless - Light yellow  
**Physical State** Liquid

**Odor** No information available  
**Odor Threshold** No data available  
**pH** > 12

**SAFETY DATA SHEET****Potassium hydroxide 1M alcoholic**

|                                                |                                               |                                             |
|------------------------------------------------|-----------------------------------------------|---------------------------------------------|
| <b>Melting Point/Range</b>                     | No data available                             |                                             |
| <b>Softening Point</b>                         | No data available                             |                                             |
| <b>Boiling Point/Range</b>                     | 78 - 80 °C / 172.4 - 176 °F                   |                                             |
| <b>Flash Point</b>                             | 15 °C / 59 °F                                 | <b>Method</b> - No information available    |
| <b>Evaporation Rate</b>                        | No data available                             |                                             |
| <b>Flammability (solid,gas)</b>                | Not applicable                                | Liquid                                      |
| <b>Explosion Limits</b>                        | <b>Lower</b> 3.3 vol%<br><b>Upper</b> 19 vol% |                                             |
| <b>Vapor Pressure</b>                          | No data available                             |                                             |
| <b>Vapor Density</b>                           | No data available                             | (Air = 1.0)                                 |
| <b>Specific Gravity / Density</b>              | 0.8                                           |                                             |
| <b>Bulk Density</b>                            | Not applicable                                | Liquid                                      |
| <b>Water Solubility</b>                        | Miscible                                      |                                             |
| <b>Solubility in other solvents</b>            | No information available                      |                                             |
| <b>Partition Coefficient (n-octanol/water)</b> |                                               |                                             |
| <b>Component</b>                               | <b>log Pow</b>                                |                                             |
| Ethyl alcohol                                  | -0.32                                         |                                             |
| Potassium hydroxide                            | 0.83                                          |                                             |
| Methyl alcohol                                 | -0.74                                         |                                             |
| <b>Autoignition Temperature</b>                | 363 °C / 685.4 °F                             |                                             |
| <b>Decomposition Temperature</b>               | No data available                             |                                             |
| <b>Viscosity</b>                               | No data available                             |                                             |
| <b>Explosive Properties</b>                    |                                               | Vapors may form explosive mixtures with air |
| <b>Oxidizing Properties</b>                    | No information available                      |                                             |

**SECTION 10. STABILITY AND REACTIVITY**

|                                         |                                                                                                                                    |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Stability</b>                        | Stable under normal conditions.                                                                                                    |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                                                      |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                           |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition.                              |
| <b>Materials to avoid</b>               | Strong oxidizing agents. Acids.                                                                                                    |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Thermal decomposition can lead to release of irritating gases and vapors. |

**SECTION 11. TOXICOLOGICAL INFORMATION****Product Information****(a) acute toxicity;**  
**Toxicology data for the components**

| Component           | LD50 Oral                                                    | LD50 Dermal                   | LC50 Inhalation                                                   |
|---------------------|--------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------|
| Ethyl alcohol       | LD50 = 10470 mg/kg<br>OECD 401 (Rat)<br>3450 mg/kg ( Mouse ) |                               | LC50 = 117-125 mg/l (4h)<br>OECD 403 (rat)<br>20000 ppm/10H (rat) |
| Potassium hydroxide | LD50 = 333-384 mg/kg (Rat)                                   |                               |                                                                   |
| Methyl alcohol      | LD50 = 1187 – 2769 mg/kg (Rat)                               | LD50 = 17100 mg/kg ( Rabbit ) | LC50 = 128.2 mg/L ( Rat ) 4 h                                     |
| Water               | -                                                            | -                             | -                                                                 |

**(b) skin corrosion/irritation;** Category 1 A

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

| Component                          | Test method                                                    | Test species | Study result    |
|------------------------------------|----------------------------------------------------------------|--------------|-----------------|
| Ethyl alcohol<br>64-17-5 ( 84.4 )  | Mouse Ear Swelling Test (MEST)                                 | mouse        | non-sensitising |
|                                    | -----<br>OECD Test Guideline 429<br>Local Lymph Node Assay     | mouse        | non-sensitising |
| Methyl alcohol<br>67-56-1 ( 3.54 ) | OECD Test Guideline 406<br>Guinea Pig Maximisation Test (GPMT) | guinea pig   | non-sensitising |

(e) germ cell mutagenicity; No data available

| Component                         | Test method                                            | Test species          | Study result |
|-----------------------------------|--------------------------------------------------------|-----------------------|--------------|
| Ethyl alcohol<br>64-17-5 ( 84.4 ) | AMES test<br>OECD Test Guideline 471                   | in vitro<br>Bacteria  | negative     |
|                                   | -----<br>Gene cell mutation<br>OECD Test Guideline 476 | in vitro<br>Mammalian | negative     |

(f) carcinogenicity;

No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

(g) reproductive toxicity; No data available

| Component                          | Test method                      | Test species / Duration       | Study result           |
|------------------------------------|----------------------------------|-------------------------------|------------------------|
| Ethyl alcohol<br>64-17-5 ( 84.4 )  | OECD Test Guideline 416          | Oral / mouse 2 Generation     | NOAEL = 13.8 g/kg/day  |
|                                    | -----<br>OECD Test Guideline 414 | Inhalation / Rat              | NOAEC = 16000 ppm      |
| Methyl alcohol<br>67-56-1 ( 3.54 ) | OECD Test Guideline 416          | Rat / Inhalation 2 Generation | NOAEC = 1.3 mg/l (air) |

(h) STOT-single exposure;

Category 2

Results / Target organs

Optic nerve  
Central nervous system (CNS)

(i) STOT-repeated exposure;

No data available

Target Organs

No information available.

(j) aspiration hazard;

No data available

Symptoms / effects, both acute and delayed

Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

## SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity effects

Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

| Component | Freshwater Fish | Water Flea | Freshwater Algae | Microtox |
|-----------|-----------------|------------|------------------|----------|
|-----------|-----------------|------------|------------------|----------|

**SAFETY DATA SHEET****Potassium hydroxide 1M alcoholic**

|                     |                                                                  |                                               |                                               |                                                                                                                       |
|---------------------|------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Ethyl alcohol       | Fathead minnow<br>(Pimephales promelas)<br>LC50 = 14200 mg/l/96h | EC50 = 9268 mg/L/48h<br>EC50 = 10800 mg/L/24h | EC50 (72h) = 275 mg/l<br>(Chlorella vulgaris) | Photobacterium<br>phosphoreum:EC50 =<br>34634 mg/L/30 min<br>Photobacterium<br>phosphoreum:EC50 =<br>35470 mg/L/5 min |
| Potassium hydroxide | LC50 = 50-165 mg/L<br>(96h)                                      |                                               |                                               |                                                                                                                       |
| Methyl alcohol      | Pimephales promelas:<br>LC50 > 10000 mg/L 96h                    | EC50 > 10000 mg/L 24h                         |                                               | EC50 = 39000 mg/L 25<br>min<br>EC50 = 40000 mg/L 15<br>min<br>EC50 = 43000 mg/L 5<br>min                              |

**Persistence and Degradability****Persistence**

Soluble in water, Persistence is unlikely, based on information available.

| Component                          | Degradability                  |
|------------------------------------|--------------------------------|
| Ethyl alcohol<br>64-17-5 ( 84.4 )  | OECD 301E = 94%                |
| Methyl alcohol<br>67-56-1 ( 3.54 ) | DT50 ~ 17.2d<br>>94% after 20d |

**Bioaccumulative Potential**

Bioaccumulation is unlikely

| Component           | log Pow | Bioconcentration factor (BCF) |
|---------------------|---------|-------------------------------|
| Ethyl alcohol       | -0.32   | No data available             |
| Potassium hydroxide | 0.83    | No data available             |
| Methyl alcohol      | -0.74   | <10 dimensionless             |

**Mobility in soil**

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

**Endocrine Disruptor Information**  
**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected endocrine disruptors

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

**SECTION 13. DISPOSAL CONSIDERATIONS****Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

|                         |                                          |
|-------------------------|------------------------------------------|
| UN-No                   | UN2924                                   |
| Proper Shipping Name    | Flammable liquid, corrosive, n.o.s.      |
| Technical Shipping Name | Contains ethanol and potassium hydroxide |
| Hazard Class            | 3                                        |
| Subsidiary Hazard Class | 8                                        |



## SAFETY DATA SHEET

Potassium hydroxide 1M alcoholic

Packing Group II

IMDG/IMO

UN-No UN2924  
Proper Shipping Name Flammable liquid, corrosive, n.o.s.  
Technical Shipping Name Contains ethanol and potassium hydroxide  
Hazard Class 3  
Subsidiary Hazard Class 8  
Packing Group II

IATA

UN-No UN2924  
Proper Shipping Name Flammable liquid, corrosive, n.o.s.  
Technical Shipping Name Contains ethanol and potassium hydroxide  
Hazard Class 3  
Subsidiary Hazard Class 8  
Packing Group II

Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component           | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|---------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Ethyl alcohol       | X                                                   | X                                       | X    | X     | 200-578-6 | X    | X   | X     | X    | X    | X    | KE-13217 |
| Potassium hydroxide | X                                                   | X                                       | X    | X     | 215-181-3 | X    | X   | X     | X    | X    | X    | KE-29139 |
| Methyl alcohol      | X                                                   | X                                       | X    | X     | 200-659-6 | X    | X   | X     | X    | X    | X    | KE-23193 |
| Water               | -                                                   | -                                       | X    | X     | 231-791-2 | X    | X   | X     | X    |      | X    | KE-35400 |

| Component      | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|----------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Methyl alcohol | 500 tonne                                                                                 | 5000 tonne                                                                               |

## National Regulations

## SECTION 16. OTHER INFORMATION

Creation Date 09-Jun-2010  
Revision Date 04-Apr-2024  
Revision Summary Not applicable.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Physical hazards**

On basis of test data

**Health Hazards**

Calculation method

**Environmental hazards**

Calculation method

### **Disclaimer**

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**End of Safety Data Sheet**